SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

Course Name: COMPUTER VISION **Course Code:** ITA05

Course Outcome (CO2): Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

CODE OF CONDUCT DECLARATION

*I, **Sriram R** (Reg. No: **192525198**), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Student Name: Sriram R **Signature:** *Sriram R*

1. Object Boundary Detection

(a) Scenario Interpretation & Importance: In medical imaging (such as MRI, CT scans, or X-rays), detecting tissue boundaries allows clinicians to locate, measure, and analyze anatomical structures (e.g., organ borders or tumor boundaries). Precise boundary detection is vital for diagnostic accuracy, surgical planning, and automated volumetric analysis.

(b) Key Challenges:

- **Low Contrast:** Biological tissues often share similar attenuation or density profiles, yielding soft, indistinct borders.
- **Noise Artifacts:** Modalities like MRI and ultrasound introduce thermal or speckle noise that distorts gradient signals.
- **Irregular Geometries:** Natural anatomical boundaries are organic and non-uniform, making simple parametric shape models unusable.

(c) Applied Method: Canny Edge Detector combined with non-maximum suppression and hysteresis thresholding.

(d) Justification: The Canny operator optimizes the trade-off between detection and localization error in low-signal-to-noise environments. It reduces high-frequency noise using Gaussian filtering, suppresses false local maxima to yield single-pixel-thin edges, and uses dual-threshold hysteresis to preserve faint boundary sections connected to strong edges.

(e) Systematic Steps:

1. **Gaussian Filter:** Smooth the image to suppress high-frequency noise.
2. **Gradient Calculation:** Compute edge magnitudes $G = \sqrt{(G_x^2 + G_y^2)}$ and directions $\theta = \arctan(G_y / G_x)$ using Sobel operators.
3. **Non-Maximum Suppression:** Thin edges by retaining only local gradient maxima.
4. **Hysteresis Thresholding:** Retain strong edge pixels and connect weak edge pixels that touch them.

2. Simple Segmentation

(a) Scenario Interpretation & Requirements: When target objects are clearly distinct from the background based on pixel intensity values, the requirement is to partition the image domain into two disjoint binary regions: foreground (objects of interest) and background.

(b) Key Enabling Features:

- **Bimodal Intensity Histogram:** Distinct peaks corresponding to background and foreground gray levels.
- **High Global Contrast:** Clear intensity separation between background and object pixels.

(c) Applied Method: Otsu's Automatic Global Thresholding Method.

(d) Justification: Otsu's algorithm automatically determines the optimal threshold T^* by maximizing inter-class variance $\sigma_B^2(T)$. Because the intensity profiles are bimodal and well-separated, Otsu's method runs efficiently with zero manual parameter tuning.

(e) Execution Flow:

1. Compute the normalized intensity histogram probabilities p_i.
2. Evaluate inter-class variance $\sigma_B^2(T) = \omega_0(T)\omega_1(T)[\mu_0(T) - \mu_1(T)]^2$ for every candidate threshold T.
3. Select $T^* = \arg\max \sigma_B^2(T)$ and classify pixels:

$$g(x, y) = 1 \quad \text{if} \quad f(x, y) \geq T^* \quad \text{else} \quad 0$$

3. Complex Segmentation

(a) Scenario Interpretation & Challenges: When multiple regions exhibit overlapping intensity values, pixel intensity alone cannot serve as a reliable discriminator. Simple global thresholding fails, resulting in heavy region misclassifications.

(b) Key Issues:

- **Intensity Overlap:** Objects and background share identical gray values due to illumination gradients, shading, or texture patterns.
- **Loss of Spatial Context:** Pixel-level thresholding ignores spatial relationships and neighborhood context.

(c) Applied Method: Marker-Controlled Watershed Segmentation.

(d) Justification: Traditional watershed causes severe oversegmentation on complex images. By introducing explicit internal (foreground) and external (background) markers, Marker-Controlled Watershed treats the gradient image as a topographic surface and floods basins specifically from these seed locations, accurately extracting boundaries despite intensity overlaps.

(e) Systematic Steps:

1. Compute the morphological gradient magnitude of the original image.
2. Extract **Internal Markers** (object seed points) and **External Markers** (background boundaries).
3. Modify the gradient image surface to contain local minima only at marker locations.
4. Perform watershed flooding algorithms to determine precise watershed boundary lines.

4. Broken Edges

(a) Scenario Interpretation: Gradient-based edge operators fail where local contrast dips below fixed thresholds or where local noise corrupts the signal, resulting in broken, discontinuous boundary segments.

(b) Key Issues:

- Discontinuous boundaries prevent shape parameterization and accurate object measurement.
- Fragmented edges break downstream topological tracking and region-growing algorithms.

(c) Applied Method: Morphological Closing combined with **Local Edge Linking (Gradient Direction Matching)** or **Hough Transform**.

(d) Justification: Morphological closing $(A \oplus B) \ominus B$ expands edge segments locally to bridge small gaps and then restores edge thickness. For structured geometrical boundaries, the Hough Transform maps edge points into parameter space to reconstruct missing boundary gaps.

(e) Operational Process:

- **Morphological Linking:** Apply dilation using a small structuring element B to bridge endpoints, followed by erosion to preserve line width.
- **Gradient Angle Linking:** Identify edge endpoints, evaluate neighboring pixels within distance d, and connect endpoints if their gradient direction angles $\theta(x, y)$ match within an angular tolerance $\Delta\theta$.

5. Combined Requirement

(a) Scenario Interpretation & Combined Goal: An industrial quality inspection system must remove unwanted noise (e.g., sensor noise) while strictly preserving fine edge structures required to identify micro-defects or surface cracks.

(b) Key Issues: Standard spatial linear filters (e.g., Mean or Gaussian filters) blur noise and sharp edge transitions indiscriminately, destroying small defect information.

(c) Applied Sequence of Techniques:

1. **Bilateral Filtering** (Edge-Preserving Noise Filter)
2. **Canny Edge Detection**

(d) Justification of Pipeline & Order:

- **Noise Removal First:** Edge operators use derivative calculations that inherently amplify high-frequency noise. Noise filtering must occur first.
- **Bilateral Filter Selection:** Unlike Gaussian filters, the Bilateral filter evaluates both domain (spatial) distance and range (intensity) differences:

$$I^{filtered}(x) = (1 / W_p) \sum I(x_i)\, g_r(\|I(x_i) - I(x)\|)\, g_s(\|x_i - x\|)$$

This smooths flat regions while preserving sharp intensity transitions.

- **Canny Edge Detection Second:** Applying Canny edge detection on the bilaterally filtered image yields clean defect boundaries without false positives caused by noise.

(e) Inspection Pipeline Overview:

Raw Inspection Image ➔ Bilateral Filtering (Edge-Preserving) ➔ Denoised Image ➔ Canny Edge Detection ➔ Defect Edge Map